

Update-Relevant Support: Hume’s Missing Descent

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Abstract

The hidden-support account of confounding gives a negative result: the same baseline public law, even augmented by the true average treatment effect, may be compatible with either zero or arbitrarily large update-relevant structural gap. This paper develops the corresponding positive theory, with one correction: arbitrary finite environment classes determine a nonempty poset of adequate supports that need not have a least element. A finite three-state example shows why the existence hypothesis is substantive. In the public-fiber-saturated case, however, the poset has a least element: the quotient generated by public observation and the fiberwise level sets of the estimand contrast. Failed public descent then has an operational measure: the largest disagreement in the target estimand between admissible environments that share a fixed public law, with a global worst-case version obtained by ranging over public laws. This is the diameter of the corresponding partial-identification set, and in saturated finite cases it has closed-form fiber-range formulae. The ordinary log-cardinality support gap remains a useful qualitative obstruction, but it controls transport risk only after calibration by effect separation and reweighting richness. The epistemological payoff is that causal warrant is not regularity plus truth; it is regularity plus the right support for update. Hume’s missing necessity becomes missing descent.

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1 Introduction

Causal inference begins from a public surface. We observe distributions of treatments, outcomes, covariates, instruments, summaries, or other measured variables, and we ask whether that surface can support a causal use: prediction under a shifted population, transport to a new setting, evaluation of an intervention, or comparison of policies. The difficulty is not only that some causes may be hidden. It is that causal use often depends on which hidden distinctions remain stable or relevant when the world is varied. A public regularity can be perfectly true at baseline while still failing to say which internal distinctions must be preserved for future projection.

The companion paper formalized this worry as a hidden-support version of Humean underdetermination [4]. It treated public observation as a projection from a richer hidden state space onto the variables actually seen. The main negative result was that one and the same baseline public law, even supplemented by the true baseline average treatment effect, can have radically different hidden reconstructions. In one reconstruction, the public variables already carry all the support needed for the local causal use. In another, the same public law and same true baseline effect conceal arbitrarily many hidden distinctions inside public fibers. Those distinctions need not change the observed baseline association, but they can become update-relevant once mass is shifted across hidden states. Public regularity, even when true and decision-correct at baseline, does not select or finitely bound the hidden support that future causal use may require.

The epistemological problem is therefore support selection: observed regularities do not determine which hidden distinctions are relevant for future use.

That negative result invites two natural complaints. First, if the support is underdetermined, relative to what standard is it underdetermined? A skeptic may grant that one can invent hidden distinctions, while denying that they were distinctions the inquiry had any reason to preserve. Second, even if a structural gap remains, why should it matter operationally? A quotient may fail to descend publicly without producing any serious disagreement in the transported estimand.

This paper answers both complaints by fixing two pieces of normative data that the baseline public law does not itself supply: an admissible class of environments $\mathcal{Q}$, saying what changes count as part of the same update problem, and an estimand ψ , saying what quantity the inquiry is trying to preserve. Once these are fixed, adequacy becomes mathematically decidable. When the adequate-support poset has a least element, that element is the update-relevant support.

Core thesis

For a fixed public projection π , admissible environment class $\mathcal{Q}$, and estimand ψ , causal adequacy is governed by the adequate-support poset induced by admissible environment variation and estimand sensitivity. When this poset has a least element, as in the public-fiber-saturated finite-linear case, that element is the update-relevant support. Failure of π to carry the relevant support induces a transport modulus: the maximum estimand movement along admissible, publicly invisible reweightings. Hume’s missing necessity becomes, in this framework, missing descent.

The central quantitative lesson is more careful than the slogan “more hidden bits imply more transport risk”. Unweighted log-cardinality does not by itself control risk. Many update-relevant hidden blocks may have arbitrarily small effect spread, while a single unresolved binary split may have a large effect contrast. The exact object is the support function of the publicly invisible

perturbation set evaluated at the estimand contrast. Cardinal gaps become risk-bearing only after calibration by effect separation, reachable reweighting, and mass budget.

The main body is intentionally finite and linear. That restriction keeps the representation theorem exact and the transport modulus transparent; nonlinear, ratio, set-valued, and continuous extensions are deferred until the finite case is in place.

Roadmap. Section 2 fixes the finite hidden-state framework, public projection, fixed response kernels, admissible environments, and linear estimands. Section 3 defines adequacy for a support relative to $(\mathcal{Q}, \psi)$. Section 4 develops the adequate-support poset and the saturated representation theorem. Section 5 proves the transport diameter/support-function identity. Section 6 separates qualitative cardinal gaps from calibrated risk gaps. Section 7 gives finite examples, including the two-state case and the additive product case. Section 8 collects the finite-linear theorem schema. Section 9 locates the account relative to transportability and sensitivity analysis. Section 10 returns to the Humean interpretation.

2 Finite fixed-kernel setup

Let $\mathcal{D}$ be a finite hidden state space and let

$$\pi : \mathcal{D} \rightarrow \mathcal{O}$$

be a public projection. A probability law q on $\mathcal{D}$ induces the public law π_*q on $\mathcal{O}$. Public observation therefore sees the pushforward and not, in general, the allocation of mass within a public fiber.

Each hidden state $u \in \mathcal{D}$ carries fixed response kernels for the interventions under consideration. In the simplest treatment-effect case, these kernels determine a state-conditional contrast

$$h(u) := \mathbb{E}[Y(1) - Y(0) \mid U = u].$$

The fixed-kernel assumption is load-bearing. Across the admissible environments studied here, what changes is the distribution over hidden states, not the response kernel attached to a hidden state. This isolates support failure from mechanism change.

Let

$$\mathcal{Q} \subseteq \Delta(\mathcal{D})$$

be the admissible class of environments. Elements of $\mathcal{Q}$ are the laws over hidden states that the update or transport problem treats as possible. The class may be generated by reweighting hidden summaries, selection mechanisms, population shifts, or other externally specified update rules. Nothing in the baseline public law alone determines $\mathcal{Q}$.

It is useful to write pushforwards as linear maps. Let

$$A_\pi : \mathbb{R}^{\mathcal{D}} \rightarrow \mathbb{R}^{\mathcal{O}}$$

be the map induced by π , so that

$$\pi_*q = A_\pi q.$$

The main text treats linear estimands

$$\psi(q) = \langle h, q \rangle := \sum_{u \in \mathcal{D}} h(u)q(u),$$

with $h : \mathcal{D} \rightarrow \mathbb{R}$. This covers the finite ATE-like case when response kernels are fixed. More complicated estimands can be locally linearized when smooth, but the exact finite theorem is clearest in the linear setting.

Definition 1 (Admissible differences). The admissible difference set generated by $\mathcal{Q}$ is

$$\text{Diff}(\mathcal{Q}) := \mathcal{Q} - \mathcal{Q} := \{q_1 - q_2 : q_1, q_2 \in \mathcal{Q}\}.$$

The publicly invisible admissible difference set is

$$\mathcal{N}_\pi(\mathcal{Q}) := \text{Diff}(\mathcal{Q}) \cap \ker A_\pi.$$

Elements of $\mathcal{N}_\pi(\mathcal{Q})$ are exactly the environment changes that can move hidden mass while leaving the public law unchanged. They are the operational version of hidden confounding for the update problem: if $z \in \mathcal{N}_\pi(\mathcal{Q})$ and $\langle h, z \rangle \neq 0$, then the estimand can change along a direction invisible to public observation.

Remark 1 (Why $\mathcal{Q}$ is normative data). The environment class is not a statistical afterthought. It specifies what kinds of reweighting, selection, or population shift the model is meant to answer to. Two inquiries may share the same public baseline law and the same estimand at baseline while differing in $\mathcal{Q}$, and therefore requiring different hidden supports.

3 Adequate supports

A support is a quotient of the hidden state space through which the public projection factors. Equivalently, it is a partition of $\mathcal{D}$ at least as fine as the public partition.

Definition 2 (Support over a public projection). A *support over π* is a surjection

$$\sigma : \mathcal{D} \twoheadrightarrow S$$

together with a map $\eta : S \rightarrow \mathcal{O}$ such that

$$\pi = \eta \circ \sigma.$$

For $q \in \Delta(\mathcal{D})$, write σ_*q for the induced distribution over support strata. Equivalently, let

$$A_\sigma : \mathbb{R}^\mathcal{D} \rightarrow \mathbb{R}^S$$

be the linear map induced by σ , so that $\sigma_*q = A_\sigma q$.

The support S records more than the public observation when a public fiber is split into several support strata. The question is whether those extra distinctions are required by the update problem.

Definition 3 ($(\mathcal{Q}, \psi)$ -adequate support). A support $\sigma : \mathcal{D} \twoheadrightarrow S$ over π is $(\mathcal{Q}, \psi)$ -adequate if there exists a function

$$\Psi_S : \sigma_*\mathcal{Q} \rightarrow \mathbb{R}$$

such that

$$\psi(q) = \Psi_S(\sigma_*q) \quad \text{for every } q \in \mathcal{Q}.$$

Thus the support is adequate when the estimand is determined by the support-stratum weights throughout the admissible environment class. Because π factors through σ , the same stratum weights also determine the public law. Adequacy therefore means that the support carries both the public side and the target estimand for the environments under consideration.

Lemma 1 (Difference criterion). For a finite linear estimand $\psi(q) = \langle h, q \rangle$, a support $\sigma : \mathcal{D} \twoheadrightarrow S$ is $(\mathcal{Q}, \psi)$ -adequate if and only if

$$\langle h, z \rangle = 0$$

for every admissible difference

$$z \in \text{Diff}(\mathcal{Q}) \cap \ker A_\sigma.$$

Proof. If $z = q_1 - q_2 \in \text{Diff}(\mathcal{Q}) \cap \ker A_\sigma$ and the support is adequate, then $\sigma_* q_1 = \sigma_* q_2$, so

$$\psi(q_1) = \Psi_S(\sigma_* q_1) = \Psi_S(\sigma_* q_2) = \psi(q_2),$$

so $\langle h, q_1 - q_2 \rangle = 0$.

Conversely, suppose $\langle h, z \rangle = 0$ for every $z \in \text{Diff}(\mathcal{Q}) \cap \ker A_\sigma$. Define Ψ_S on $\sigma_* \mathcal{Q}$ by choosing any $q \in \mathcal{Q}$ with $s = \sigma_* q$ and setting

$$\Psi_S(s) := \psi(q).$$

If q_1 and q_2 are two choices with the same support pushforward, the assumption gives $\psi(q_1) = \psi(q_2)$. Hence Ψ_S is well defined and the support is adequate. $\square$

Lemma 1 is the finite linear core of the sequel. It says that a support is adequate exactly when it kills all estimand movement along admissible differences that it cannot see.

4 Representation of update-relevant support

The base hidden-support framework starts with a target family $\mathcal{E}$ and forms the least quotient through which the public projection and every target invariant factor [3]. In the present paper the target family is not posited independently. It is induced by $(\mathcal{Q}, \psi)$.

Definition 4 (Unresolved estimand-sensitive direction). A support $\sigma : \mathcal{D} \rightarrow S$ over π has an *unresolved estimand-sensitive direction* if there exist $q_1, q_2 \in \mathcal{Q}$ such that

$$\sigma_* q_1 = \sigma_* q_2 \quad \text{but} \quad \psi(q_1) \neq \psi(q_2).$$

Equivalently, there exists

$$z \in \text{Diff}(\mathcal{Q}) \cap \ker A_\sigma$$

with $\langle h, z \rangle \neq 0$.

By Lemma 1, unresolved estimand-sensitive directions are exactly the admissible differences inside the support kernel that are not annihilated by h .

Definition 5 (Minimal and least adequate supports). An adequate support is *minimal* if no strictly coarser support is adequate. A *least* adequate support is an adequate support that is coarser than every adequate support. Finite problems always have minimal adequate supports, because the discrete support is adequate and the partition lattice is finite. They need not have a least adequate support.

Definition 6 (Update-relevant support). An *update-relevant support* for $(\pi, \mathcal{Q}, \psi)$ is the least $(\mathcal{Q}, \psi)$ -adequate support over π , whenever such a least support exists.

Proposition 1 (Finite existence criterion). Let $\mathcal{D}$ be finite, and let $\mathcal{A}(\mathcal{Q}, \psi)$ be the nonempty set of $(\mathcal{Q}, \psi)$ -adequate supports over π . Let σ_0 be the common coarsening of all supports in $\mathcal{A}(\mathcal{Q}, \psi)$, equivalently the partition whose equivalence relation is generated by the union of their equivalence relations. Then a least adequate support exists if and only if σ_0 is adequate. In that case, σ_0 is the least adequate support.

Proof. If a least adequate support exists, it is coarser than every adequate support and is therefore the common coarsening of all adequate supports. Hence it is σ_0 , and σ_0 is adequate. Conversely, if σ_0 is adequate, then by construction it is coarser than every adequate support. It is therefore least. $\square$

Example 1 (The least adequate support need not exist). Let

$$\mathcal{D} = \{a, b, c\}$$

lie in a single public fiber, let

$$h(a) = 0, \quad h(b) = 1, \quad h(c) = 2,$$

and let

$$\mathcal{Q} = \left\{ \left(\frac{1}{3}, \frac{1}{3} + t, \frac{1}{3} - t \right) : |t| \leq \frac{1}{3} \right\}.$$

The only admissible difference direction is a scalar multiple of $\delta_b - \delta_c$, and

$$\langle h, \delta_b - \delta_c \rangle = -1 \neq 0.$$

By Lemma 1, a support is adequate exactly when it separates b from c . The two partitions

$$\{\{a, b\}, \{c\}\} \quad \text{and} \quad \{\{a, c\}, \{b\}\}$$

are both minimal adequate supports, but they are incomparable. Their common coarsening is the trivial partition, which merges b with c and is inadequate. Thus a least adequate support need not exist. The example also shows why an unrestricted formula that simply joins public observation with all estimand-sensitive distinctions can be too fine: the discrete partition is adequate here, but it is not least.

The example explains why the representation theorem below is stated under a real richness hypothesis rather than for arbitrary $\mathcal{Q}$. In the finite reweighting settings targeted by this paper, the clean base case is public-fiber saturation: once the public marginal is fixed, every redistribution inside a public fiber is admissible.

Definition 7 (Public-fiber-saturated reweighting class). Let $P \subseteq \Delta(\mathcal{O})$ be a convex polytope of admissible public marginals. The associated *public-fiber-saturated* class is

$$\mathcal{Q}_P := \{q \in \Delta(\mathcal{D}) : \pi_* q \in P\}.$$

Equivalently, after an admissible public marginal $p \in P$ is chosen, the conditional law inside each fiber $\pi^{-1}(o)$ may be any distribution on that fiber. We assume, after discarding irrelevant states if necessary, that every public fiber under discussion is active: for each $o \in \text{im}(\pi)$ there exists $p \in P$ with $p(o) > 0$.

This hypothesis is strong, but it is crisp. It is also exactly the case needed for the closed-form transport formulae below. More restricted reweighting classes require a lattice analysis of adequate supports; Example 1 shows that existence and meet-closure cannot be assumed.

Definition 8 (Fiberwise estimand quotient). Define the quotient $\pi_h : \mathcal{D} \rightarrow \mathcal{D}/\sim_{\pi, h}$ by

$$u \sim_{\pi, h} v \iff \pi(u) = \pi(v) \text{ and } h(u) = h(v).$$

Equivalently, π_h is the join of the public partition with the level-set partition of h .

Lemma 2 (Fiberwise decomposition). Let $\mathcal{Q}_P$ be public-fiber-saturated. If $z = q_1 - q_2$ with $q_1, q_2 \in \mathcal{Q}_P$ and $\pi_* q_1 = \pi_* q_2$, then

$$z = \sum_{o \in \text{im}(\pi)} z_o,$$

where each z_o is supported on $\pi^{-1}(o)$ and satisfies

$$\sum_{u \in \pi^{-1}(o)} z_o(u) = 0.$$

Conversely, any sufficiently small zero-sum transfer inside a public fiber is realized as an admissible difference between two elements of $\mathcal{Q}_P$ with the same public marginal, provided the starting law assigns positive mass to the negative part of the transfer.

Proof. If $\pi_* q_1 = \pi_* q_2$, then the signed measure $z = q_1 - q_2$ has total mass zero on each public fiber. Decomposing z by public fibers gives the displayed sum. Conversely, choose an admissible law $q \in \mathcal{Q}_P$ with positive mass on every state in the support of the negative part of z_o . For sufficiently small $\varepsilon > 0$, the law $q + \varepsilon z_o$ remains nonnegative and has the same public marginal as q , hence both laws lie in $\mathcal{Q}_P$. $\square$

Theorem 1 (Representation under public-fiber saturation). Assume $\mathcal{D}$ is finite, response kernels are fixed across $\mathcal{Q}_P$, and $\psi(q) = \langle h, q \rangle$ is linear. Then

$$\mathcal{U}(\mathcal{Q}_P, \psi) = \pi_h$$

is the least $(\mathcal{Q}_P, \psi)$ -adequate support over π .

Proof. First, π_h is adequate. If two laws $q_1, q_2 \in \mathcal{Q}_P$ have the same π_h -pushforward, then they assign the same total mass to every joint value of (π, h) . Since h is constant on each π_h -block,

$$\psi(q_1) = \sum_{B \in \mathcal{D}/\sim_{\pi, h}} h(B) (\pi_h)_* q_1(B) = \sum_{B \in \mathcal{D}/\sim_{\pi, h}} h(B) (\pi_h)_* q_2(B) = \psi(q_2).$$

So Lemma 1 gives adequacy.

Conversely, let $\sigma : \mathcal{D} \rightarrow S$ be any adequate support over π . Since it is a support over π , it cannot merge states with different public images. Suppose it merges u and v with

$$\pi(u) = \pi(v) \quad \text{but} \quad h(u) \neq h(v).$$

Choose $p \in P$ with $p(\pi(u)) > 0$, and choose $q \in \mathcal{Q}_P$ whose conditional law on $\pi^{-1}(\pi(u))$ gives positive mass to both u and v . For sufficiently small $\varepsilon > 0$, define

$$q^+ := q + \varepsilon(\delta_u - \delta_v), \quad q^- := q - \varepsilon(\delta_u - \delta_v).$$

Both laws lie in $\mathcal{Q}_P$, and

$$\pi_* q^+ = \pi_* q^-$$

because u and v lie in the same public fiber. Also

$$\sigma_* q^+ = \sigma_* q^-$$

because u and v lie in the same σ -block. But

$$\psi(q^+) - \psi(q^-) = 2\varepsilon(h(u) - h(v)) \neq 0.$$

This contradicts Lemma 1. Hence every adequate support must separate all states separated by π_h , so every adequate support refines π_h . Therefore π_h is the least adequate support. $\square$

Corollary 1 (Public descent). Under the hypotheses of Theorem 1, public observation is adequate for $(\mathcal{Q}_P, \psi)$ if and only if

$$\mathcal{U}(\mathcal{Q}_P, \psi) = \pi.$$

Equivalently, every estimand-sensitive admissible direction already descends through the public projection.

Proof. If $\mathcal{U}(\mathcal{Q}_P, \psi) = \pi$, the public law itself carries the least adequate support. Conversely, if the public law is adequate, then the coarsest adequate support cannot be finer than π , while every support over π refines π . Hence the coarsest adequate support is π . $\square$

Proposition 2 (Monotonicity of support). Let $\mathcal{Q}' \subseteq \mathcal{Q}$, and suppose least adequate supports exist for both $(\mathcal{Q}, \psi)$ and $(\mathcal{Q}', \psi)$. Then $\mathcal{U}(\mathcal{Q}, \psi)$ refines $\mathcal{U}(\mathcal{Q}', \psi)$.

Proof. Since $\mathcal{Q}' \subseteq \mathcal{Q}$, every support adequate for $\mathcal{Q}$ is adequate for $\mathcal{Q}'$. Therefore the least adequate support for $\mathcal{Q}'$, if it exists, is at least as coarse as the least adequate support for $\mathcal{Q}$. Equivalently, $\mathcal{U}(\mathcal{Q}, \psi)$ refines $\mathcal{U}(\mathcal{Q}', \psi)$. $\square$

5 Transport moduli and support functions

The representation results identify the adequate-support poset for a specified update problem, and its least element in the saturated case. The transport modulus measures what remains when the public projection does not carry the distinctions adequacy requires.

Definition 9 (Local and global transport moduli). For an observed public law $p \in \pi_* \mathcal{Q}$, define the *local transport modulus*

$$\Omega_{\mathcal{Q}}(p; \psi) := \sup\{|\psi(q_1) - \psi(q_2)| : q_1, q_2 \in \mathcal{Q}, \pi_* q_1 = \pi_* q_2 = p\}.$$

This is the diameter of the partial-identification set

$$\{\psi(q) : q \in \mathcal{Q}, \pi_* q = p\}.$$

The *global transport modulus* is

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) := \sup_{p \in \pi_* \mathcal{Q}} \Omega_{\mathcal{Q}}(p; \psi).$$

The local modulus is attached to a fixed observed public law. The global modulus records the worst-case ambiguity over public laws realized by the admissible environment class.

Theorem 2 (Transport diameter/support-function identity). If $\mathcal{D}$ is finite and $\psi(q) = \langle h, q \rangle$, then

$$\Omega_{\mathcal{Q}}(p; \psi) = \sup\{|\langle h, z \rangle| : z = q_1 - q_2, q_i \in \mathcal{Q}, A_{\pi} q_i = p\}.$$

Consequently,

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) = \sup_{z \in \mathcal{N}_{\pi}(\mathcal{Q})} |\langle h, z \rangle|.$$

Proof. For the local identity, every admissible pair with public law p gives a difference $z = q_1 - q_2$ satisfying $A_{\pi} q_1 = A_{\pi} q_2 = p$, and

$$|\psi(q_1) - \psi(q_2)| = |\langle h, q_1 - q_2 \rangle| = |\langle h, z \rangle|.$$

Taking the supremum over the same admissible pairs gives the equality. Taking the supremum of the local identity over $p \in \pi_* \mathcal{Q}$ gives the global identity, since the union of the local difference sets is exactly $\mathcal{N}_{\pi}(\mathcal{Q})$. $\square$

Theorem 2 identifies the operational object. In partial-identification language, $\Omega_{\mathcal{Q}}(p; \psi)$ is the diameter of the set of ψ -values compatible with public law p and $\mathcal{Q}$ [7]. The present framework links that diameter to quotient structure: failed public descent matters exactly when the public kernel of the environment class contains directions not annihilated by the estimand. Since $\mathcal{N}_{\pi}(\mathcal{Q})$ is centrally symmetric, the global modulus is the support function of the publicly invisible admissible difference set, evaluated at h .

Proposition 3 (Closed forms under public-fiber saturation). Let $\mathcal{Q}_P = \{q \in \Delta(\mathcal{D}) : \pi_* q \in P\}$. For each public value o , define the fiber range

$$R_h(o) := \max_{u \in \pi^{-1}(o)} h(u) - \min_{u \in \pi^{-1}(o)} h(u).$$

For $p \in P$,

$$\Omega_{\mathcal{Q}_P}(p; \psi) = \sum_{o \in \text{im}(\pi)} p(o) R_h(o).$$

Consequently,

$$\Omega_{\mathcal{Q}_P}^{\text{glob}}(\pi; \psi) = \max_{p \in P} \sum_{o \in \text{im}(\pi)} p(o) R_h(o).$$

In particular, if $P = \Delta(\mathcal{O})$, then

$$\Omega_{\mathcal{Q}_P}^{\text{glob}}(\pi; \psi) = \max_{o \in \text{im}(\pi)} R_h(o),$$

and if $P = \{p\}$, then

$$\Omega_{\mathcal{Q}_P}^{\text{glob}}(\pi; \psi) = \sum_{o \in \text{im}(\pi)} p(o) R_h(o).$$

Proof. For any pair $q_1, q_2 \in \mathcal{Q}_P$ with common public marginal p , the difference in conditional expected h inside each fiber is at most $R_h(o)$. Hence

$$|\psi(q_1) - \psi(q_2)| \leq \sum_o p(o) R_h(o).$$

This bound is attained, for fixed p , by putting q_1 on maximizers of h in each fiber and q_2 on minimizers. This proves the local formula. Maximizing the local formula over $p \in P$ gives the global formula and its two stated special cases. $\square$

Remark 2 (Estimand contrast modulo public functions). For any function $\varphi : \mathcal{O} \rightarrow \mathbb{R}$, replacing h by

$$h + \varphi \circ \pi$$

leaves $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi)$ unchanged, because

$$\langle \varphi \circ \pi, z \rangle = 0 \quad \text{for every } z \in \ker A_{\pi}.$$

Thus transport ambiguity depends only on the class of h modulo publicly measurable functions. Under public-fiber saturation, this class is measured by the fiberwise ranges $R_h(o)$.

Corollary 2 (Public adequacy and zero modulus). For finite linear estimands,

$$\pi \text{ is } (\mathcal{Q}, \psi)\text{-adequate} \iff \Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) = 0.$$

For a fixed public law p , $\Omega_{\mathcal{Q}}(p; \psi) = 0$ says only that ψ is identified at p within $\mathcal{Q}$.

Proof. The public projection π is $(\mathcal{Q}, \psi)$ -adequate exactly when every $z \in \text{Diff}(\mathcal{Q}) \cap \ker A_\pi$ is annihilated by h , by Lemma 1 applied to $\sigma = \pi$. By Theorem 2, this is equivalent to $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) = 0$. The local statement follows directly from the definition of $\Omega_{\mathcal{Q}}(p; \psi)$. $\square$

Proposition 4 (Monotonicity of transport ambiguity). If $\mathcal{Q}' \subseteq \mathcal{Q}$, then

$$\Omega_{\mathcal{Q}'}^{\text{glob}}(\pi; \psi) \leq \Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi).$$

Proof. The supremum defining $\Omega_{\mathcal{Q}'}^{\text{glob}}$ ranges over a subset of the pairs used to define $\Omega_{\mathcal{Q}}^{\text{glob}}$. $\square$

Remark 3 (Why this is stronger than a scalar bias check). A baseline bias calculation evaluates one environment or one family of perturbations around it. The local modulus $\Omega_{\mathcal{Q}}(p; \psi)$ ranges over all admissible environments with the same public law p , while the global modulus ranges over all such public laws. These are structural sensitivity parameters for the update problem, not reports of present aggregate bias.

6 Cardinal gaps and calibration

The companion paper defines a structural confounding gap as a log-cardinality: how many minimally adequate hidden support blocks may remain unresolved inside one public fiber. That quantity remains useful here, but it must be handled carefully.

Definition 10 (Update-relevant cardinal gaps). Assume $\mathcal{U}(\mathcal{Q}, \psi)$ is finite. For $o \in \text{im}(\pi)$, let

$$m_{\pi, \mathcal{Q}, \psi}(o)$$

be the number of $\mathcal{U}(\mathcal{Q}, \psi)$ -blocks contained in $\pi^{-1}(o)$. The maximum-fiber cardinal gap is

$$\Gamma_{\max}(\pi; \mathcal{Q}, \psi) := \max_{o \in \text{im}(\pi)} \log_2 m_{\pi, \mathcal{Q}, \psi}(o).$$

The additive total cardinal gap is

$$\Gamma_{\text{tot}}(\pi; \mathcal{Q}, \psi) := \sum_{o \in \text{im}(\pi)} \log_2 m_{\pi, \mathcal{Q}, \psi}(o).$$

When no qualifier is given, Γ_{card} denotes $\Gamma_{\max}$.

The cardinal gap vanishes exactly when the update-relevant support descends publicly. But it does not, by itself, determine the magnitude of transport risk.

Proposition 5 (Uncalibrated cardinality does not control risk). There is no strictly increasing function $c : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ with $c(0) = 0$ such that, for all finite linear update problems,

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) \geq c(\Gamma_{\text{card}}(\pi; \mathcal{Q}, \psi)).$$

Nor is there a finite function C of Γ_{card} alone that upper-bounds $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi)$ for all such problems.

Proof. For the lower-bound failure, take a single public fiber containing 2^n update-relevant states and let $\mathcal{Q}$ permit arbitrary reweighting inside the fiber. Set $h(u) \in [0, \varepsilon]$ with distinct values. Then $\Gamma_{\text{card}} = n$, but

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) \leq \varepsilon.$$

Since ε is arbitrary, no positive lower bound depending only on n can hold.

For the upper-bound failure, take a single public fiber with two hidden states a, b , let $\mathcal{Q}$ permit moving mass between them, and set $h(a) - h(b) = M$. Then $\Gamma_{\text{card}} = 1$, while $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi)$ can be made as large as M . $\square$

The right quantitative object is therefore an effect-weighted and reachability-weighted gap. In the linear finite case, this object is already the transport modulus.

Definition 11 (Budgeted risk gap). For a symmetric admissible budget set $B \subseteq \mathcal{N}_\pi(\mathcal{Q})$, define

$$\Gamma_{\text{risk}}^B(\pi; \mathcal{Q}, \psi) := \sup_{z \in B} |\langle h, z \rangle|.$$

When $B = \mathcal{N}_\pi(\mathcal{Q})$, this is exactly $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi)$.

Corollary 3 (Risk gap equals the transport modulus). For finite linear estimands,

$$\Gamma_{\text{risk}}^{\mathcal{N}_\pi(\mathcal{Q})}(\pi; \mathcal{Q}, \psi) = \Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi).$$

Proof. This is Theorem 2 with notation changed to emphasize the risk-calibrated reading. $\square$

The unweighted cardinal gap still has a calibrated use. It lower-bounds risk when the hidden distinctions are not only numerous, but also quantitatively separated and reachable by admissible reweighting.

Theorem 3 (Calibrated cardinal lower bound). Suppose a public fiber contains $m = 2^\Gamma$ update-relevant blocks $B_1, \dots, B_m$ ordered so that their estimand levels satisfy

$$\inf_{u \in B_{i+1}} h(u) - \sup_{u \in B_i} h(u) \geq \alpha > 0$$

for $i = 1, \dots, m-1$. Suppose there exist $q_-, q_+ \in \mathcal{Q}$ with $\pi_* q_- = \pi_* q_+$ such that

$$q_+ - q_- = \rho(\nu_m - \nu_1),$$

where $\rho > 0$, ν_1 is a probability distribution supported on B_1 , and ν_m is a probability distribution supported on B_m . Then

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) \geq \rho \alpha (2^\Gamma - 1).$$

Proof. By the separation assumption, any $u_1 \in B_1$ and $u_m \in B_m$ satisfy

$$h(u_m) - h(u_1) \geq \alpha(m-1).$$

Let $z = q_+ - q_-$. By hypothesis, $z = \rho(\nu_m - \nu_1)$ is publicly invisible. Therefore

$$|\langle h, z \rangle| \geq \rho \alpha (m-1) = \rho \alpha (2^\Gamma - 1).$$

Theorem 2 gives the same lower bound for $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi)$. $\square$

Remark 4 (Saturation by bounded range). If ψ has a bounded range, then the hypotheses of Theorem 3 cannot force a lower bound larger than the realized range of ψ over publicly equivalent admissible environments. In examples where mass-transfer budgets increase, the lower bound grows only until the available range is exhausted.

Proposition 6 (Independent calibrated directions). Suppose there exist publicly invisible admissible differences

$$z_1, \dots, z_k \in \mathcal{N}_\pi(\mathcal{Q})$$

such that $|\langle h, z_i \rangle| \geq \beta_i > 0$. Suppose also that $\mathcal{Q}$ permits their simultaneous composition with signs chosen to align their contributions: for some $\epsilon_i \in \{-1, 1\}$,

$$z = \sum_{i=1}^k \epsilon_i z_i \in \mathcal{N}_\pi(\mathcal{Q})$$

and $\epsilon_i \langle h, z_i \rangle \geq \beta_i$ for every i . Then

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) \geq \sum_{i=1}^k \beta_i.$$

Proof. By construction,

$$\langle h, z \rangle = \sum_{i=1}^k \epsilon_i \langle h, z_i \rangle \geq \sum_{i=1}^k \beta_i.$$

Since $z \in \mathcal{N}_{\pi}(\mathcal{Q})$, the claim follows from Theorem 2. $\square$

Remark 5 (The moral of calibration). Hidden structure is not dangerous merely because it is numerous. It becomes decision-relevant when admissible environment variation can move mass across distinctions the estimand cares about by a non-negligible amount.

7 Worked finite examples

Example 2 (One unresolved binary split). Let one public fiber contain two hidden states a and b , so $\pi(a) = \pi(b) = o$. Let $\mathcal{Q}$ permit arbitrary reweighting between a and b while keeping the public mass on o fixed. Let

$$h(a) - h(b) = \alpha.$$

Then $\mathcal{U}(\mathcal{Q}, \psi)$ separates a from b , while π does not. Hence

$$\Gamma_{\max}(\pi; \mathcal{Q}, \psi) = 1.$$

The two extreme environments, all mass on a and all mass on b , have the same public law and differ in ψ by $|\alpha|$. Therefore

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) = |\alpha|.$$

The cardinal gap says that one public observation leaves one update-relevant bit unresolved; the transport modulus says how much that bit can move the estimand.

Example 3 (Many blocks with tiny effect spread). Let a single public fiber contain 2^n hidden states and let $\mathcal{Q}$ permit arbitrary reweighting inside the fiber. Assign distinct estimand levels in the interval $[0, \varepsilon]$. Then the update-relevant cardinal gap is n , but the transport modulus is at most ε . This is the basic reason that uncalibrated log-cardinality cannot lower-bound transport risk.

Example 4 (A single split with large effect contrast). Let the public fiber contain two hidden states a, b , and let $h(a) - h(b) = M$. Then the cardinal gap is only one bit, but $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) = M$ when full transfer is admissible. This is the complementary reason that cardinality alone cannot upper-bound transport risk.

Example 5 (Additive product systems). Take k independent copies of the binary example, with contrasts α_i and independent publicly invisible reweighting budgets. Suppose the product construction creates one public fiber containing all 2^k hidden product states and the estimand is additive across coordinates. If the subset sums

$$\sum_{i \in I} \alpha_i \quad (I \subseteq \{1, \dots, k\})$$

are all distinct, then the saturated update-relevant quotient has 2^k blocks and

$$\Gamma_{\max} = k$$

while, under full product-fiber reweighting,

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) = \sum_{i=1}^k |\alpha_i|$$

until any bounded-range saturation applies. If the subset sums are not all distinct, then

$$\Gamma_{\max} = \log_2 N,$$

where N is the number of distinct subset sums. For example, if all $\alpha_i = \alpha$, then $N = k + 1$ rather than 2^k .

If instead the k binary gaps live in separate public fibers and the gap is defined as a maximum over fibers, the displayed cardinal gap remains one bit per fiber. The additive burden is instead tracked by the total-gap convention

$$\Gamma_{\text{tot}} := \sum_{i=1}^k \Gamma_i.$$

This distinction matters because maximum-fiber cardinality and compositional hidden-support burden answer different questions.

8 Finite-linear theorem schema

The preceding results form a finite-linear schema, split between the arbitrary finite case, where adequacy gives a poset, and the public-fiber-saturated case, where a least support is guaranteed.

Theorem 4 (Finite-linear update-support schema). Let $\mathcal{D}$ be finite, let $\pi : \mathcal{D} \rightarrow \mathcal{O}$ be public observation, let $\mathcal{Q} \subseteq \Delta(\mathcal{D})$ be an admissible environment class, and let $\psi(q) = \langle h, q \rangle$ be a linear estimand. For any support σ over π , let A_σ be the associated pushforward map. Then:

- (i) σ is $(\mathcal{Q}, \psi)$ -adequate if and only if

$$\langle h, z \rangle = 0 \quad \text{for every } z \in \text{Diff}(\mathcal{Q}) \cap \ker A_\sigma.$$

- (ii) For each public law $p \in \pi_* \mathcal{Q}$,

$$\Omega_{\mathcal{Q}}(p; \psi) = \sup\{|\langle h, z \rangle| : z = q_1 - q_2, \ q_i \in \mathcal{Q}, \ A_\pi q_i = p\},$$

and

$$\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) = \sup_{z \in \text{Diff}(\mathcal{Q}) \cap \ker A_\pi} |\langle h, z \rangle|.$$

- (iii) Public observation π is $(\mathcal{Q}, \psi)$ -adequate if and only if $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi) = 0$. By contrast, $\Omega_{\mathcal{Q}}(p; \psi) = 0$ only says that ψ is identified at the particular public law p .
- (iv) The adequate supports form a nonempty finite poset. A least adequate support exists if and only if the common coarsening of all adequate supports is itself adequate.
- (v) If $\mathcal{Q} = \mathcal{Q}_P$ is public-fiber-saturated, then the least adequate support exists and is the quotient

$$\pi_h : u \mapsto (\pi(u), h(u)).$$

In that case, for $p \in P$,

$$\Omega_{\mathcal{Q}_P}(p; \psi) = \sum_{o \in \text{im}(\pi)} p(o) R_h(o),$$

and the global modulus is obtained by maximizing this expression over $p \in P$.

Proof. Clause (i) is Lemma 1. Clause (ii) is Theorem 2. Clause (iii) is Corollary 2. Clause (iv) follows from finiteness of the partition lattice and Proposition 1. Clause (v) combines Theorem 1 and Proposition 3. $\square$

9 Relation to existing causal methodology

The framework is adjacent to transportability theory, but it asks a prior question. Pearl–Bareinboim-style transportability starts with assumptions about shared or shifted mechanisms and asks whether an effect can be transported or identified [2, 8]. The present account asks which hidden distinctions a specified update task makes adequate, and when those distinctions collapse to a least quotient. If measured variables fail to realize them, the ambiguity is measured locally by $\Omega_{\mathcal{Q}}(p; \psi)$ and globally by $\Omega_{\mathcal{Q}}^{\text{glob}}(\pi; \psi)$.

The framework is also adjacent to invariance-based causal learning. Invariant causal prediction and invariant risk minimization treat stability across environments as evidentially central [1, 9]. The update-support account agrees, but separates two levels: given an environment family and estimand, one can ask which support is adequate; the family and estimand are not themselves recovered from the baseline observational surface.

Finally, the account is related to sensitivity analysis. Rosenbaum-style methods and related robustness analyses ask how strong unmeasured confounding would need to be to change a conclusion [10]. The present modulus is a structural cousin, but it asks how far the estimand can move across all admissible environments that remain publicly indistinguishable, not under one parametric bias model.

Positioning

Existing transportability methods ask whether transport succeeds given a selection structure. Sensitivity analysis asks how robust a conclusion is under a chosen hidden-bias model. Update-relevant support asks which hidden distinctions the update task itself makes adequate, and the transport modulus measures the ambiguity left when those distinctions do not descend publicly.

10 The Humean payoff

The companion paper’s Humean point was that regularities plus a correct baseline effect do not fix hidden support. In Hume’s terms, inspection does not reveal the power by which an effect must follow, and repeated conjunction does not by itself justify projection [6, T 1.3.6, SBN 86–94]; [5, E 4–5, SBN 25–55]. This paper sharpens that point into a three-part epistemological claim.

Public evidence does not select its own support. The same public regularity can be compatible with different hidden-support structures, and those structures can disagree under update, intervention, or transport. The missing element is not simply an unobserved cause, but a principle licensing some hidden distinctions as relevant while compressing others away.

Causal knowledge is task-relative but not arbitrary. The finite theory explains what becomes determinate after one fixes what could have varied and what quantity the inquiry targets. In the present notation, $\mathcal{Q}$ and ψ determine an adequacy standard and a finite poset of adequate supports; under additional richness hypotheses, they determine a least support. Public regularity alone is insufficient because adequacy is constituted by a task and an admissible space of variation. The

right causal representation is intrinsic not to the observed regularity alone, but to that regularity under a target question and counterfactual environment class.

Hume’s gap becomes measurable. The transport diameter identity explains when failed descent matters. The residual gap is not metaphysical residue. It matters exactly when publicly invisible admissible differences move the estimand:

$$z \in \mathcal{N}_\pi(\mathcal{Q}) \quad \text{and} \quad \langle h, z \rangle \neq 0.$$

That condition is the finite-linear form of update-relevant hidden confounding.

The Humean gap is the failure of the public projection to carry the update-relevant quotient, when such a least quotient exists, and more generally the distinctions demanded by the adequate-support poset. Habit, repetition, and baseline correctness close the gap only if they reveal the distinctions needed for transport; otherwise the residual is measured by publicly invisible directions along which the estimand can move.

Missing descent

Causal warrant is not regularity plus truth; it is regularity plus the right support for update. The epistemic unit of causal knowledge is not an observed association, nor even a true effect, but a public projection that descends through the quotient required for the intended counterfactual use.

The calibration results prevent overstatement. A large hidden-support cardinality gap is not automatically a large decision risk; danger requires public invisibility, admissible reachability, and estimand separation. Humean underdetermination is real, but its operational size is measured by the specified update problem rather than by raw hidden multiplicity.

The resulting position is neither skeptical nor narrowly empiricist. Observation does not determine the hidden support required for causal adequacy, but a legitimate environment class and estimand make the adequate-support poset and transport ambiguity exact; in saturated finite-linear cases, they also identify the least required support.

11 Extensions and limits

The finite linear case is the right base case, but three extensions are natural.

First, smooth nonlinear estimands can be treated locally by replacing h with the relevant influence function. The update-sensitive quotient then depends on the local environment and may change across $\mathcal{Q}$. Ratio estimands, standardized effects, and policy values fall under this heading when differentiability conditions hold.

Second, set-valued or nonsmooth estimands require replacing influence functions with subdifferentials or correspondences. The representation theorem then becomes one-sided: a support that preserves every active subgradient is sufficient, but minimality may require additional regularity assumptions.

Third, continuous hidden spaces require replacing log-cardinality by entropy, metric entropy, dimension, or another measure of fiber complexity. The transport modulus still has the same conceptual form:

$$\sup\{|\psi(q_1) - \psi(q_2)| : \pi_*q_1 = \pi_*q_2, \ q_1, q_2 \in \mathcal{Q}\},$$

but existence and compactness assumptions become substantive.

The fixed-kernel assumption is also essential. If response kernels may change across environments, then ψ can move because mechanisms changed, not because hidden support was reweighted. The clean equivalence between hidden-support failure and transport ambiguity then breaks into separate support-change and mechanism-change components.

12 Conclusion

Hidden support becomes determinate only relative to an update problem, and even then determinacy is not automatic. Once $\mathcal{Q}$ and ψ are fixed, arbitrary finite problems determine a nonempty poset of adequate supports. A least support exists exactly when the common coarsening of those supports remains adequate; in saturated finite-linear reweighting classes, it is the quotient generated by public observation and the fiberwise level sets of the estimand contrast.

When public observation is inadequate, the operational residue is not raw hidden cardinality. It is the transport modulus: the maximum estimand disagreement between admissible, publicly indistinguishable environments. In the finite linear case this modulus is the support function of the publicly invisible perturbation set against the estimand contrast.

This is the two-sided sequel to the hidden-support/Humean-underdetermination result. The negative theorem says that baseline public data cannot select or finitely bound the update-relevant support gap. The positive theory says what the task selects: an adequate-support poset, a least support in the saturated cases, and an exact measure of the residual transport risk.

The epistemological payoff is that causal knowledge requires more than true public regularity, but less than metaphysical access to hidden necessity. Public observation supplies a projection of the hidden causal state; causal use requires a task-relative adequacy standard generated by admissible variation and the estimand to be transported. The gap between the two is Humean underdetermination in finite-linear form: public evidence may fail to determine the support on which future causal update depends, but that failure can be represented, measured, and bounded.

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